Project: Worked examples
Title: Bolt group (elastic method)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A pattern of bolts in a plane under in-plane forces (with eccentricity, so torsion), out-of-plane moments and axial load: the shear and tension in every bolt by the elastic (area-weighted) method, the governing bolt, and an optional lever-model prying addition.

WHAT IT CALCULATES	RELATION
Centroid of the bolt group, each bolt weighted by its area	$x_c = \Sigma A\,x / \Sigma A$ (y_c likewise)
Torsion about the centroid from the eccentric load	$T = F_y\,(e_x - x_c) - F_x\,(e_y - y_c) + T_z$
Polar moment of the bolt areas about the centroid	$J = \Sigma A\,r^2$
Shear on each bolt, its direct share plus the torsion (vector sum)	$v = A\,(F/\Sigma A) + T{\cdot}A{\cdot}r/J$
Tension on each bolt, its direct share plus the moments, with the product of inertia so that an unsymmetric pattern is right	$t = A\,N/\Sigma A + A{\cdot}[(M_x\,I_y - M_y\,I_{xy})\,y' + (M_y\,I_x - M_x\,I_{xy})\,x']/(I_x\,I_y - I_{xy}^2)$, where $I_x = \Sigma A\,y'^2$, $I_y = \Sigma A\,x'^2$, $I_{xy} = \Sigma A\,x'y'$ (this reduces to $M_x\,A\,y'/\Sigma A\,y'^2 + M_y\,A\,x'/\Sigma A\,x'^2$ only for a pattern symmetric about a centroidal axis)
Tension with prying, when the distances a and b are given	$t_p = t\,(1 + b/a)$
Combined shear and tension, circular interaction (elastic method, Shigley-style)	$(v/V_R)^2 + (t_p/T_R)^2 \le 1$
Combined shear and tension, EN 1993-1-8 linear form (each is also checked alone; the highest governs)	$v/V_R + t_p/(1.4\,T_R) \le 1$

Basis: Design resistance of individual fasteners; elastic distribution in a bolt group[1](§3.6, §3.8); Eccentrically loaded bolt groups — elastic method[2](Part 7).

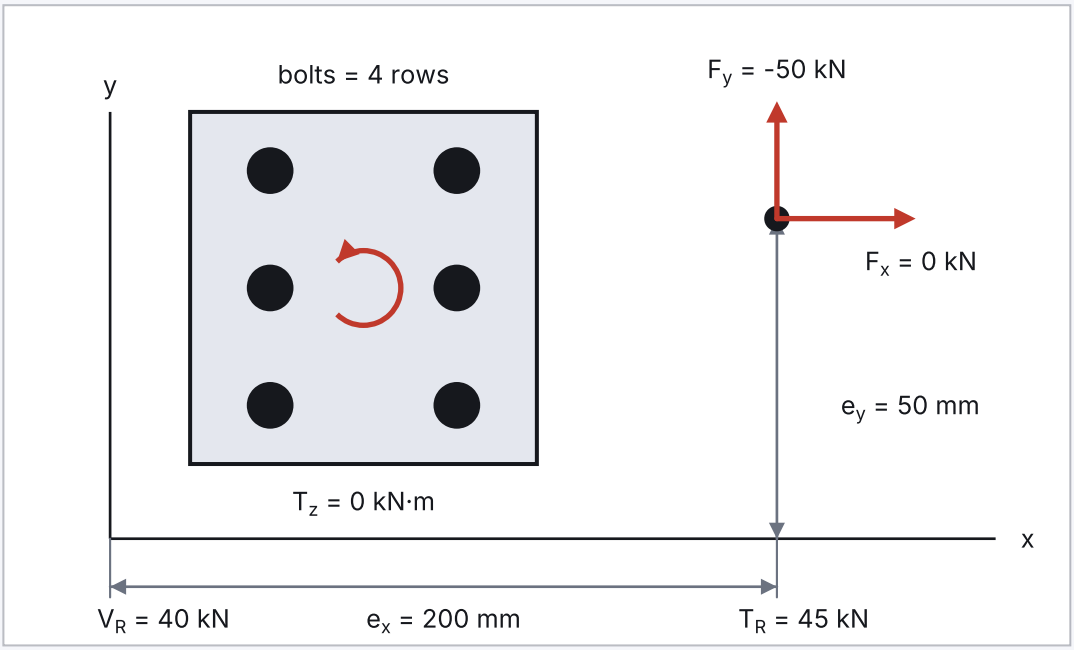


Figure 1 — A bolt group (six drawn; the bolts as listed) under an in-plane load at (e_x, e_y), x and y from one origin, + as drawn. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
bolts	Bolt positions and areas	4 row(s)		drawing: DRG-BRK-09 rev B	any
F_x	In-plane force, x	0	kN	spec: LOAD-CASE-1	-1000000 … 1000000 kN
F_y	In-plane force, y	-50	kN	spec: LOAD-CASE-1	-1000000 … 1000000 kN
e_x	Point of application, x	200	mm	drawing: DRG-BRK-09 rev B	-100000 … 100000 mm
e_y	Point of application, y	50	mm	drawing: DRG-BRK-09 rev B	-100000 … 100000 mm
T_z	Additional in-plane torque about the centroid	0	kN·m	spec: LOAD-CASE-1	-1000000 … 1000000 kN·m
N	Axial (out-of-plane) load, tension positive	0	kN	spec: LOAD-CASE-1	-1000000 … 1000000 kN
M_x	Out-of-plane moment about x (tension on +y side)	0	kN·m	spec: LOAD-CASE-1	-1000000 … 1000000 kN·m
M_y	Out-of-plane moment about y (tension on +x side)	0	kN·m	spec: LOAD-CASE-1	-1000000 … 1000000 kN·m
V_R	Shear resistance per bolt	40	kN	spec: M12 8.8 shear resistance, EN 1993-1-8 Table 3.4 per project calc	0.1 … 10000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T_R	Tension resistance per bolt	45	kN	spec: M12 8.8 tension resistance per project calc	0.1 … 10000 kN

TABLE ROW	VALUES
bolts[1]	x = 0 mm; y = 0 mm; A = 84.3 mm²
bolts[2]	x = 100 mm; y = 0 mm; A = 84.3 mm²
bolts[3]	x = 100 mm; y = 100 mm; A = 84.3 mm²
bolts[4]	x = 0 mm; y = 100 mm; A = 84.3 mm²

3. Assumptions

A1. The connected parts are rigid; bolt forces vary linearly with distance from the centroid (elastic method). (Conservative compared with plastic / instantaneous-centre methods.)
A2. Resistances are per bolt as entered, with their source (a design-code capacity or a manufacturer value). (No code capacity is computed here.)
A3. Prying uses a simple lever: the bolt tension rises by the factor (1 + b/a). (An optional, conservative first estimate.)

4. Working

[11] n_b = number of bolts
= 4
= 4
Bolts in the group
[12] $x_c = \Sigma A\ x\ /\ \Sigma A$
= 84.3×0 + 84.3×100 + 84.3×100 + 84.3×0 / 337.2
= 50 mm
Centroid x
[13] $y_c = \Sigma A\ y\ /\ \Sigma A$
= 84.3×0 + 84.3×0 + 84.3×100 + 84.3×100 / 337.2
= 50 mm
Centroid y
[14] $T = F_y\ (e_x - x_c) - F_x\ (e_y - y_c) + T_z$
= -50 × 150 − 0 × 0 + 0
= -7.5 kN·m
In-plane torque about the centroid
[15] $J = \Sigma A\ r^2$
= about the centroid
= 1686000 mm⁴
Polar second moment of the bolt areas
[16] b1 = bolt 1 at (0, 0) mm
= v = (-18.75, 6.25) kN, t = 0 kN
= 0.49411
Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)}$ = 0.494
[17] b2 = bolt 2 at (100, 0) mm
= v = (-18.75, -31.25) kN, t = 0 kN
= 0.91109
Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)}$ = 0.911
[18] b3 = bolt 3 at (100, 100) mm
= v = (18.75, -31.25) kN, t = 0 kN
= 0.91109
Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)}$ = 0.911

[19] b4 = bolt 4 at (0, 100) mm
= v = (18.75, 6.25) kN, t = 0 kN
= 0.49411
Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)}$ = 0.494
[20] v_{max} = max over the bolts of v
= bolt 2
= 36.443 kN
Maximum shear on any bolt
[21] t_{max} = max over the bolts of t (with prying)
= bolt 1
= 0 kN
Maximum tension on any bolt
[22] $bolt_v$ = bolt with the largest shear
= bolt 2
= "bolt 2"
The bolt carrying the largest shear
[23] $bolt_t$ = bolt with the largest tension
= bolt 1
= "bolt 1"
The bolt carrying the largest tension
[24] $u_{max} = \sqrt{((v/V_R)^2 + (t/T_R)^2)}$
$= \sqrt{((36.443/40)^2 + (0/45)^2)}$
= 0.91109
Governing interaction utilisation
[25] governing = governing bolt
= bolt 2
= "bolt 2"
The bolt with the highest utilisation
[26] $u_v = v_{max} / V_R$
= 36.443 kN / 40 kN
= 0.91109
Shear utilisation of the most sheared bolt
[27] $u_t = t_{max} / T_R$
= 0 kN / 45 kN
= 0
Tension utilisation of the most tensioned bolt
[28] $governing_{lin}$ = bolt with the highest linear utilisation
= bolt 2
= "bolt 2"
The bolt that governs the linear shear–tension check
[29] u_{lin} = max over the bolts of $v/V_R + t/(1.4\ T_R)$
= bolt 2: 36.443/40 + 0/(1.4 × 45)
= 0.91109
Linear shear–tension interaction, the highest over all bolts

5. Results

SYMBOL	VALUE	DESCRIPTION
T	-7.500 kN·m	Torque about the centroid
J	1686000 mm⁴	Polar second moment of bolt areas
v_{max}	36.44 kN	Maximum shear on any bolt
t_{max}	0 kN	Maximum tension on any bolt
$bolt_v$	"bolt 2"	Bolt with the largest shear

SYMBOL	VALUE	DESCRIPTION
$bolt_t$	"bolt 1"	Bolt with the largest tension
governing	"bolt 2"	Governing bolt
$governing_{lin}$	"bolt 2"	Bolt governing the linear interaction
u_{max}	0.91	Governing interaction utilisation (governing)
u_{lin}	0.91	Linear interaction utilisation (highest over all bolts)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Circular shear–tension interaction (governing bolt)	u_{max} = 0.91	u_{max} ≤ 1	0.91	
PASS	Linear shear–tension interaction v/V_R + t/(1.4 T_R), every bolt	u_{lin} = 0.91	u_{lin} ≤ 1	0.91	[1](Table 3.4)
PASS	Shear alone (largest bolt shear)	u_v = 0.91	u_v ≤ 1	0.91	
PASS	Tension alone (largest bolt tension)	u_t = 0.00	u_t ≤ 1	0.00	

7. Applicability

None.

8. Verification

Template mech.bolt-group@1.4.0 Content hash 51d1bceeeb8e14ba Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. American Institute of Steel Construction. Steel Construction Manual. 15th ed. AISC; 2017.